

International Islamic University Chittagong
Center for General Education(CGED)

Midterm Examination, Spring-26

Course Code: GEED-2302(GEED- 2101)

Course Title: Sciences of *Qur'an* and *Hadith*

Time: 1:30 Hours

Full Marks: 30

Answer the following questions
(All questions are of equal value)

SL	Questions	Marks	CLOs	Bloom's taxonomy domain
1	Explain the definition of the holy <i>Qur'an</i> literally and terminologically. Summarize some names of the holy <i>Qur'an</i> proving it to be the speech of Allah. Or, Explain the necessity of the holy <i>Qur'an</i> in human life.	10	2	Create
2	Define <i>Wahi</i> literally and terminologically. Explain its different types.	10	2	Remember & Create
3	Define <i>Ayah</i> and <i>Surah</i> of the holy <i>Qur'an</i> . Evaluate the opinions of scholars regarding the arrangement of <i>Surah</i> in the holy <i>Qur'an</i> .	10	2	Remember & Evaluate

International Islamic University Chittagong

Department of Computer Science & Engineering

Program: B.Sc. in CSE: Semester: 3rd

Mid Term Special Examination, Spring-2026

Course Code: CSE-2323

Course Title: Digital Logic Design

Time: 1 Hour 30 minutes.

Total Marks: 30

Answer the following Three (3) questions. Each question carries 10 marks.

Question No	Question	CLOs	Marks
1.a	I. A 4-bit data word $D = 1011$ is to be transmitted using Hamming code. Construct the Hamming code word using even parity II. Suppose the received code is 1010011. Detect whether an error has occurred. If yes, locate the error bit position. Correct the code and determine the original data word.	CLO1	5
1.b	Is the BCD code a self-complementing code? Justify your answer. Perform the following 4-digit BCD addition: $3758 + 4867$ Express the final result in valid BCD form.	CLO1	3
1.c	Given a decimal digit 8, perform the following conversions: I. Convert the decimal number into its Excess-3 code II. Determine the corresponding Gray code.	CLO1	2
2.a	Simplify the Boolean function with don't care condition using K-Map: $F(w, x, y, z) = w'(x'y + x'y' + xyz) + x'z'(y + w)$ $d(w, x, y, z) = w'x(y'z + yz) + wy$	CLO1	4
2.b	Given the Boolean function $F(A, B, C, D) = B'D + A'CD + ABCD$ Perform the following tasks: I. Convert the expression into Sum-of-Products (SOP) form II. Draw the truth Table III. Simplify (F) using a Karnaugh map and provide the minimized expression.	CLO1	6
	OR		

2.b	Simplify the Boolean function using K-map. $F(A, B, C, D, E) = \sum (0, 2, 3, 6, 7, 8, 10, 11, 16, 18, 20, 22, 25, 26, 27, 28)$	CLO1	6																																				
3.a	Design a combinational circuit that converts a 3-bit binary number into its 1's complement. Show truth table and logic implementation	CLO2	5																																				
	OR																																						
3.a.	Design a 1-bit full adder combinational circuit using only NOR gates and analyze its gate count and logic depth.	CLO2	5																																				
3.b	I. after simplification, Implement the function $F(X, Y, Z)$ using AOI (AND-OR-Inverter) logic II. Convert the AOI implementation into an equivalent NAND-only circuit	CLO1	5																																				
	<table> <tr> <th>X</th><th>Y</th><th>Z</th><th>F</th></tr> <tr><td>0</td><td>0</td><td>0</td><td>0</td></tr> <tr><td>0</td><td>0</td><td>1</td><td>0</td></tr> <tr><td>0</td><td>1</td><td>0</td><td>0</td></tr> <tr><td>0</td><td>1</td><td>1</td><td>1</td></tr> <tr><td>1</td><td>0</td><td>0</td><td>0</td></tr> <tr><td>1</td><td>0</td><td>1</td><td>1</td></tr> <tr><td>1</td><td>1</td><td>0</td><td>1</td></tr> <tr><td>1</td><td>1</td><td>1</td><td>1</td></tr> </table>	X	Y	Z	F	0	0	0	0	0	0	1	0	0	1	0	0	0	1	1	1	1	0	0	0	1	0	1	1	1	1	0	1	1	1	1	1		
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International Islamic University Chittagong
Department of Computer Science and Engineering
B.Sc. in CSE Midterm Examination, Spring 2026
Course Code: STAT 2311 Course Title: Probability and Statistics
Total marks: 30, Time: 1 hour 30 minutes

[Answer all the questions; Figures in the right-hand margin indicate full marks.]

CO DL

- 1(a) Define Statistics. Analyze the relationship between statistics and computer science. 5 **CO1 C4**
- 1(b) Differentiate between probability and non-probability sampling. A researcher wants to study the opinion of 10,000 students. Which sampling method is most suitable and why? 5 **CO1 C5**

OR

Compare primary and secondary data in terms of reliability, accuracy, cost, and time. A population consists of 4 departments in IIUC: CSE, EEE, BBA, and Law. Suggest a suitable sampling technique to ensure representation from each department and justify your choice.

- 2(a) Define geometric mean and harmonic mean. Why are measures of central tendency important in data analysis? 5 **CO1 C2**

- 2(b) The following data shows the number of hours spent on coding practice by 50 CSE students per day: 5 **CO1 C3**

Hours	0-2	2-4	4-6	6-8	8-10
Frequency	6	10	14	12	8

- (i) Compute the mean and median hours of coding practice.
(ii) Draw the ogive curve from the above data.

- 3(a) Differentiate between variance and standard deviation. Evaluate the significance of coefficient of variation in comparing two datasets. 5 **CO1 C5**

- 3(b) CPU usage (%) of two servers measured over six-time intervals is given below: 5 **CO1 C4**

Interval	1	2	3	4	5	6
Server-A	60	70	65	75	65	68
Server-B	66	67	65	68	66	67

- (i) Which server has higher average usage?
(ii) Which server shows more consistency?

OR

The response time (in ms) of a web server is recorded as:
2, 5, 7, 9, 11

CO1 C4

- (i) Verify the inequality: $A.M \geq G.M \geq H.M$
(ii) Compute σ_x and $MD_{\bar{x}}$ and verify: $\sigma_x > MD_{\bar{x}}$

International Islamic University Chittagong (IIUC)
Department of Computer Science and Engineering
Mid Term Examination

Program: **B.sc (Eng.)**
 Course Code: **CHEM-2301**
 Total Marks: **30**

Semester: **Spring- 2026**
 Course Title: **Chemistry**
 Time: **1 Hour 30 Minutes**

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
 (ii) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions

- CLO1** Developing the basic knowledge of different chemicals.
CLO2 Understanding of the chemical behaviour and physical properties of common substances.
CLO3 Demonstrate the basic proficiency to solve the chemical problems and discuss the interactions between matter and energy at the atomic and molecular levels.

Bloom's Levels of the Questions

Letter Symbols
Meaning

R
Remember

U
Understand

Ap
Apply

An
Analyze

E
Evaluate

C
Create

1.	(a)	Define Periodic Law. Distinguish between the Classical and Modern Periodic Tables.	R	CLO1	5
	(b)	Define Pauli's Exclusion Principle and demonstrate its application using the outermost shell electrons of a Sulfur atom.	U	CLO2	5
2.	(a)	Define Ionization Energy (I.E.) and discuss its periodic trends across periods and groups with appropriate examples.	E	CLO3	5
	(b)	Define Electron Affinity (E.A.) and explain why Fluorine has a lower Electron Affinity than Chlorine.	An	CLO2	5
	OR				
	(c)	Define Chemical Bonding. Predict and explain the type of bond formed between Aluminum and Chlorine with the help of a diagram.	Ap	CLO1	5
	(d)	What is a metallic bond? Explain how metallic bonding leads to electrical conductivity.	An	CLO2	5
3.	(a)	What are the differences between Transition and Inner Transition elements?	R	CLO1	5
	(b)	Why are transition and inner transition elements important in modern technology and computer engineering? Explain with examples.	C	CLO3	5

Time: 1:30 hours

Marks: 30

[Answer all questions; Read the instruction below before answering the questions; Figures in right hand margin indicate full marks]

1. a) If $\begin{bmatrix} 3 & -1+x & 2 \\ 3 & -1 & x+2 \\ x+3 & -1 & 2 \end{bmatrix}$ is singular and $x \in [-4, -1]$, what is the value of x 3 CLO1 C2

Or

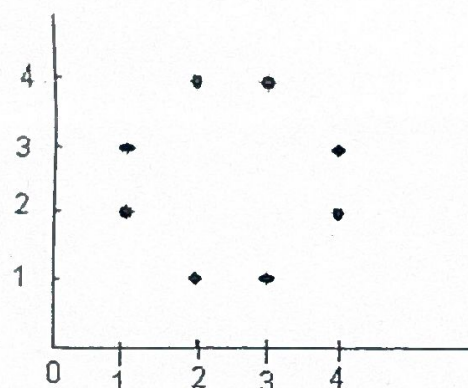
Test the given vectors are orthonormal

$$\vec{u} = [2/5, 1/5, -2/5, 4/5] \text{ and } \vec{v} = [3/\sqrt{65}, -6/\sqrt{65}, 4/\sqrt{65}, 2/\sqrt{65}]$$

3 CLO1 C2

- b) Find Inverse of the matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ by making Linear Equations 3 CLO1 C2

c) "



2 CLO1 C2

Create an adjacency matrix for the above relation

- d) What is the relation between Matrix, Eigen Vector and Eigen value? 2 CLO1 C2
 When do we use free variables when solving linear systems of equations? Give an example.

Or

Test whether $\lambda_1 = f$ and $\lambda_2 = e$ are Eigen values for $A = \begin{bmatrix} 5 & 7 \\ 2 & 4 \end{bmatrix}$;

2 CLO2 C2

Where e and f are the digits of your own ID number

2. a) Determine which of the following vectors are Eigen vectors for 5 CLO2 C2

$$A = \begin{pmatrix} 2 & f \\ -1 & -5 \end{pmatrix} \text{ showing your analysis procedure graphically}$$

$$(a) \begin{pmatrix} b \\ f \end{pmatrix} \quad (b) \begin{pmatrix} b \\ e \end{pmatrix}; \text{ Where } b, e \text{ and } f \text{ are the digits of your own ID number}$$

b) Determine whether the set of matrices

$$\left\{ \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 1 \\ 0 & 1 \end{bmatrix} \right\} \text{ is linearly independent.}$$

Or

Verify Cayley Hamilton theorem for the matrix $\begin{bmatrix} 2 & -1 \\ -3 & 4 \end{bmatrix}$ and hence find A^{-1} .

3.

Diagonalize the matrix $\begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$

Instruction:

If your ID no is C171023 then you can assign your digit as follows

C	1	7	1	0	2	3
	↓	↓	↓	↓	↓	↓
	a	b	c	d	e	f

To solve the above problems, you need to use your own ID number where needed

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International Islamic University Chittagong
 Department of Computer Science & Engineering
Mid Term Examination, Spring 2026
CSE 2321 Data Structures
 Total marks: 30 Time: 90 minutes

[Answer *all* of the following questions. Figures in the right-hand margin indicate full marks.]

- 1.a) A library maintains a book list containing the following data for each book: 2 CO1
 Name of Authors, Title of Book, Edition, Publisher's Name, Year, ISBN C2
 i) State the *entities*, *attributes* and *entity set* of the list.
 ii) Which attribute can serve as the *primary key* for the list? }
- b) Draw a *flowchart* for the *binary search* algorithm. 3 CO1
 [OR] C2
 Draw a flowchart to remove all duplicate elements from a sorted array in-place (without using extra memory) and return the new length of the array.
- c) What do you mean by *complexity of algorithms*? What is the *time complexity* of the following functions? Explain. 2 CO3
C4
 i)

```
int func1(int n){
    int i = 1, count = 0;
    while(i < n){
        i = i * 3;
        count++;
    }
    return count;
}
```

 ii)

```
int func2(int n){
    int count = 0;
    for(int i = 0; i < n; i++){
        for(int j = n; j > 0; j = j / 2){
            count++;
        }
    }
    return count;
}
```
- d) For the following pattern P and text T, find the number C of comparisons to find the INDEX of P in T using the pattern matching algorithm you studied. You have to show each step. P = WXXW, T = WXWXXWXXXWXXW. 3 CO3
C3
- 2.a) What is *linear array*? How can we represent a linear array in memory? 2 CO1
 [OR] C2
 How a 2D array is represented in computer memory? Explain with example.
- b) Consider a 2D array A (10 : 15, 13 : 20). 2.5 CO3
 i) Find the length of each dimension and the number of elements in A. C3
 ii) Suppose Base (A) = 200 and w = 4 words per memory cell for A.
 Find the address of the element A [13, 17] in *row-major order*.
- c) Given a string ROBOT, find the number of *comparisons* (C) and number of *interchanges* (D) needed to *sort* the string *alphabetically* by using *bubble sort* algorithm. Show each steps. 4 CO3
 [OR] C3
 Show the steps to search the ITEM = 41 and ITEM = 79 from the following list of integers using *binary search* algorithm.
 10, 15, 22, 28, 34, 41, 47, 56, 63, 70, 78, 85, 92
- d) Write down the *worst case* & *average case* time complexity for the following algorithm: 1.5 CO1
 i) Binary search ii) Linear search iii) Bubble Sort C1

3.a) What is *linked list*? What is the difference between an array and a linked list in terms of their Memory allocation, Insertion and deletion? Which one is more efficient for searching an element, and why?

2 C01
C1

b) You are given only a pointer to a node (not the head pointer) in a non-empty singly linked list. Write a C/C++ function to delete that node from the list in $O(1)$ time. Under what condition does your approach fail, and why?

4 C03
C3

[OR]

You are given a two-way linked list containing an odd number of elements. You don't know how many numbers there are in that list. Write an algorithm or C/C++ code to find the middle element of this list by iterating only once over the list. What is the time complexity of this algorithm?

c) Given is the following sorted linked list where the array INFO contains a list of integer numbers and LINK, START and AVAIL are the pointer fields-

2 C01
C2

	1	2	3	4	5	6	7	8	9	10	11	12
INFO:			41	129		9	123	78	194	231	62	145
LINK:	2	5	11	12	0	3	4	7	10	0	8	9

START : 6 AVAIL : 1

Determine the changes in the list if number XX is *added* to the list and then 123 is *deleted* from the list.

[Here XX is the last two digits of your ID]

d) What is *two way linked list*? Form a *two way linked list* from the one way list in Q 3(c).

2 C01
C2